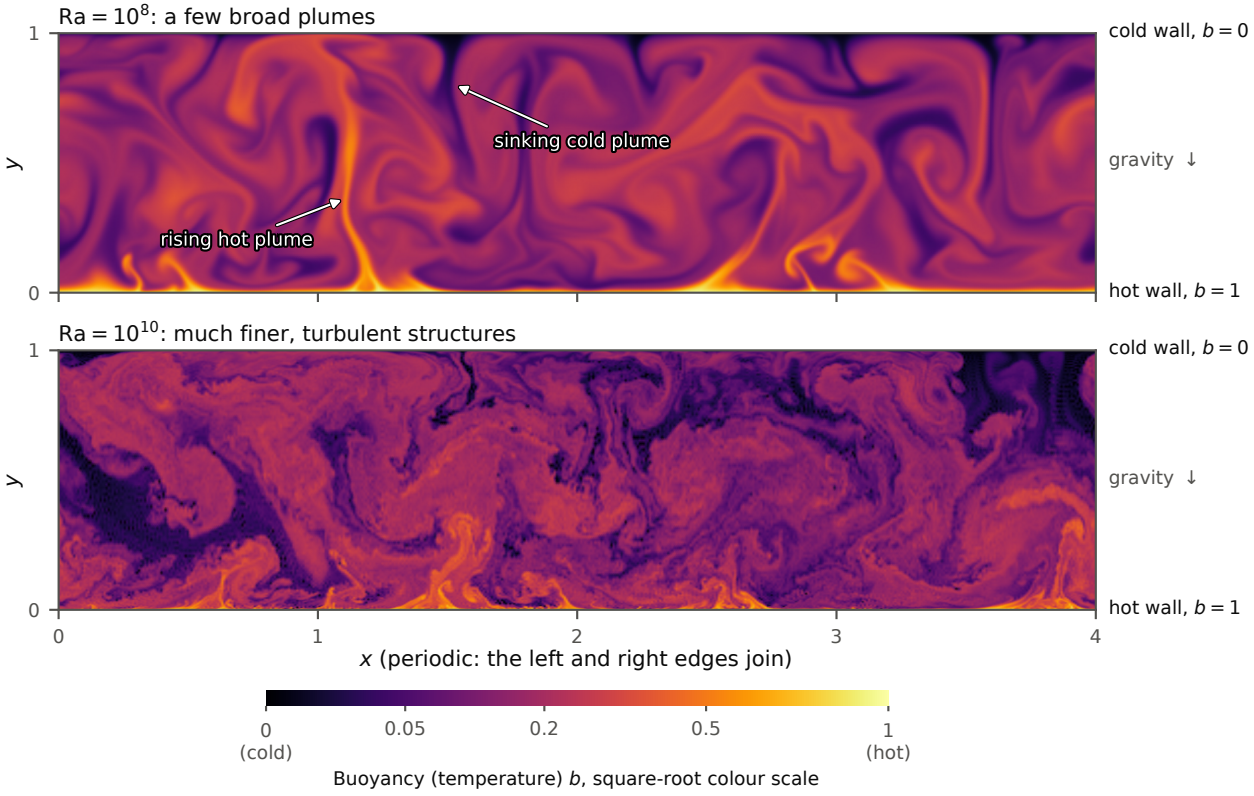

Ra = 10^8: a few broad plumes
cold wall, b = 0
gravity ↓
hot wall, b = 1
sinking cold plume
rising hot plume
y
1
0
Ra = 10^10: much finer, turbulent structures
cold wall, b = 0
gravity ↓
hot wall, b = 1
y
1
0
0
1
2
3
4
x (periodic: the left and right edges join)
0
(cold)
0.05
0.2
0.5
1
(hot)
Buoyancy (temperature) b, square-root colour scale